

Initial Pebble Model Friction Imbalance Model – Summary

Model Idea:

The icesheet is covered in small pebbles that are created when water is sprayed on the ice before the game starts, without these pebbles, there is no curl. I am assuming the pebbles are randomly distributed across the ice, with a constant radius and height. To determine a value for the number of pebbles that are on the icesheet, Figure 1 is used.

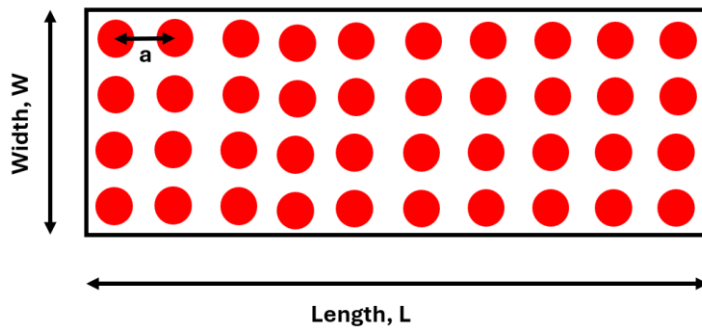
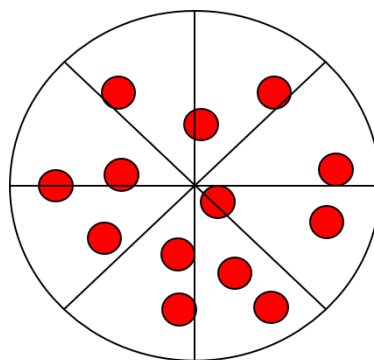


Figure 1: A diagram showing how I have estimated the number of pebbles on the ice sheet

To get an estimate for the number of pebbles, n , that are on the ice, it is just length, l in metres, of the ice divided by the average pebble separation, a in metres, multiplied by width, w in metres, divided by the average pebble separation. This is shown by the below equation.

$$n = \frac{l \times w}{a^2}$$

This model relies on the change in friction between different parts of the stone caused by different contact areas touching each section of the stone. To model this, I have split the stone into 8 angular sections, then calculate the surface area per angular section to find a coefficient of friction value per angular section.



Pebble Imbalance Model

Figure 2: A diagram showing the curling stone split into 8 angular sections, with the randomly distributed pebbles shown in red

In this model, I am accounting for only the velocity and contact area as factors that affect the coefficient of friction value.

Then, the difference between the coefficient of friction value between opposite sections is calculated and the force is calculated using the equation $F = \mu N$ where F is the difference in force, μ is the coefficient of friction and N is the normal force. Then, the forces are summed in the x and y directions.

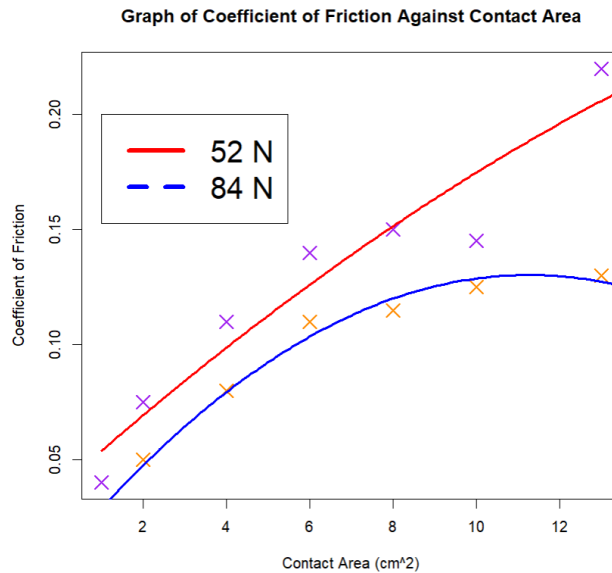


Figure 3: A diagram showing coefficient of friction versus contact area curves

The equation I have determined, using the equations of the curves of best fit from figure 3 and the fact that the coefficient of friction is proportional to the velocity over 2, for the coefficient of friction is

$$\mu = \left(\frac{v}{2}\right) \times \left(\frac{m}{84} \times 7.89 \times 10^{-3} + 0.019x - 6000x^2\right)$$

The data points for the graph in figure 3 are from a published source on ice friction (Kietzig et al., 2010).

Optimisation:

I am splitting the icesheet into multiple grids so that when the stone is checking how many pebbles the running band is in contact with, it only has to check the grid cell the stone is in and the 8 adjacent cells. This is done so that the model does not take as long to run.

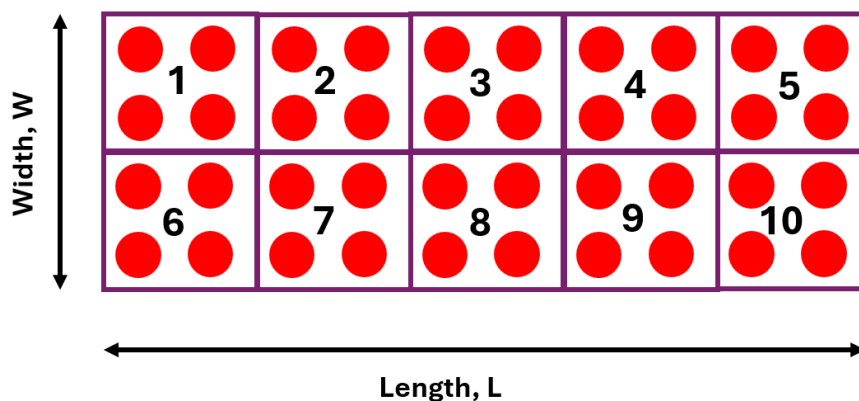


Figure 4: A diagram showing how the pebbles are split up into grids on the icesheet

Assumptions:

I am assuming that the pebbles are randomly distributed with a fixed height and radius. I am also assuming that there is always the same number of pebbles on the ice as well as every pebble having the same contact area. I am also assuming that only the angular section that includes the centre of a pebble is in contact with it. This model also assumes that the only factors affecting the coefficient of friction are

Predictions:

I predict that there will be no overall curl produced by this model, this is because the pebbles on the ice are randomly distributed and therefore all forces should cancel out.

Results:

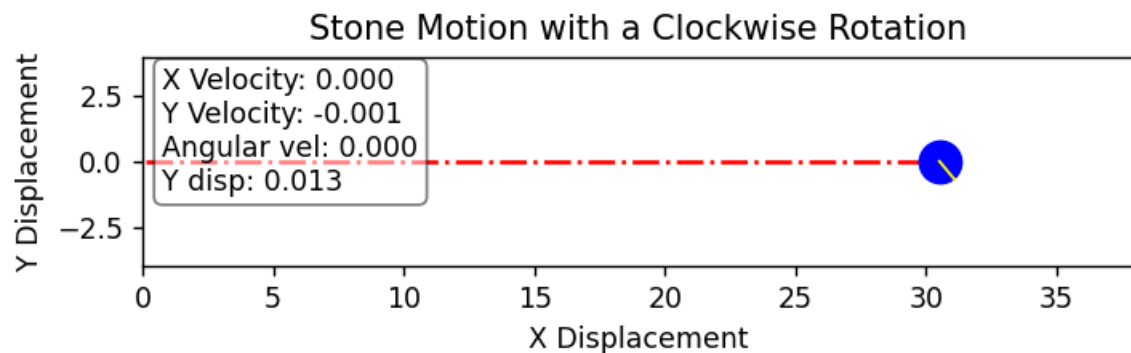


Figure 5: A diagram showing the curling stone at the end of its path with a displacement in the y direction of 13cm

Conclusion:

Overall, I predicted that the stone would not curl due to the pebbles being randomly distributed leading to the forces in the y direction balancing out. The results achieved by the model also has a very small curl in the y direction showing that the prediction is accurate.

Future Work:

- 1) Include the height of the pebbles as well as differing radiuses and contact areas
- 2) Account for thermal effects
- 3) Include the effects of scratches on the ice